



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2008
HIGHER SCHOOL CERTIFICATE
ASSESSMENT TASK #3

Mathematics Extension 1

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators may be used.
- All necessary working should be shown in every question.
- Hand in your answers in 3 separate bundles. Section A (Question 1), Section B (Question 2) and Section C (Question 3)

Total Marks – 80

- Attempt questions 1-3
- All questions are **NOT** of equal value.

Examiner: *R. Boros*

This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

Section A – Start a new booklet.**Question 1 (27 marks).****Marks****a)** Differentiate with respect to x

(i) $\frac{x}{2} \ln(x^2 - 2)$ 2

(ii) $\cos^{-1}(3x - 1)$ 2

b) If M and N are constants and $x = Me^{-t} + Ne^{2t}$ simplify:

$$\frac{d^2x}{dt^2} - \frac{dx}{dt} - 2x$$
 4

c) In how many ways can eight different books be arranged on a row such that three particular books are always together? 2

d) Find

(i) $\int \frac{1 dx}{(4 + x)^2}$

(ii) $\int \frac{1 dx}{4 + x^2}$

(iii) $\int \frac{x dx}{4 + x^2}$ 6

e) Use the substitution $u = x - 1$ to evaluate:

$$\int_0^1 x(x - 1)^5 dx$$
 4

f) What is the largest possible domain of:

$$y = \ln(x^2 - 1)$$
 3

g) Four digit numbers greater than 7000 are to be made from the digits 8, 7, 6 and 5. Repetition of the digits is allowed. How many such numbers can be made? 2

h) (i) How many different arrangements of the letters of BALLOON are possible? 1

(ii) How many of these start with the letter L. 1

End of Section A.

Section B – Start a new booklet.**Question 2 (29 Marks).****Marks**

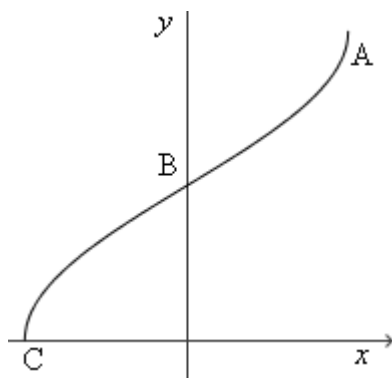
a) Consider the function $f(x) = 2 \tan^{-1}\left(\frac{x}{2}\right)$.

- | | | |
|-------|-----------------------------|---|
| (i) | Evaluate $f(0)$. | 1 |
| (ii) | Draw the graph of $f(x)$. | 1 |
| (iii) | State the domain and range. | 2 |

b) For the function $y = x + e^{-x}$

- | | | |
|-------|--|---|
| (i) | Find the coordinates and determine the nature of any stationary points on the graph of $y = f(x)$. | 3 |
| (ii) | Show that the graph is always concave upwards for all values of x . | 1 |
| (iii) | Sketch the graph of $y = f(x)$ showing clearly the coordinates of any stationary points and the equations of any asymptotes. | 2 |

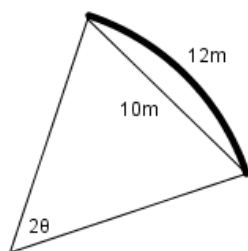
c) The diagram below shows the graph of $y = \pi + 2 \sin^{-1} 3x$.



- | | | |
|-------|---|---|
| (i) | Write down the coordinates of the point B. | 2 |
| (ii) | Write down the coordinates of the endpoints A and C. | 4 |
| (iii) | Find the equation of the tangent to the curve $y = \pi + 2 \sin^{-1} 3x$ at the point B (give answer in general form of equation) | 3 |

Section B continues next page

- d) A pipe which is 12 metres long is bent into a circular arc which subtends an angle of 2θ radians at the centre of the circle. The chord of the circle joining the ends of the arc is 10 metres long.



- | | | |
|-------|---|---|
| (i) | Show that $6\sin\theta - 5\theta = 0$. | 3 |
| (ii) | Show that $\theta_0 = 1$ is a good first approximation to the value of θ . | 2 |
| (iii) | Use one application of Newton's Method to find another approximation, θ_1 , to the value of θ (correct to 4 decimal places). | 3 |
| (iv) | Use this value of θ_1 , to find an approximation to the length of the radius of the arc, rounding off this answer correct to 2 decimal places. | 2 |

End of Section B.

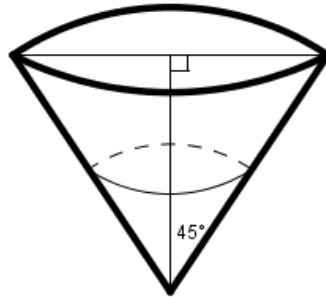
Section C – Start a new booklet.**Question 3 (24 marks).****Marks****a)**

Figure not to scale.

Water is being let into the conical vessel, shown above, at a constant rate of $8\text{cm}^3/\text{second}$.

- (i) Show that if h cm is the depth of the vessel then $r = h$ and

$$V = \frac{1}{3}\pi h^3 \quad 2$$

When the depth is 12cm find:

- (ii) The rate of increase in the depth (in terms of π). 3
 (iii) The rate of increase in the area of the top surface of the water. 3

- b)** The rate of decrease of temperature of a body hotter than its surroundings is proportional to the temperature difference. If A is the air temperature and T is the temperature of the body after t minutes, then

$$\frac{dT}{dt} = -k(T - A) \quad (1)$$

- (i) Show that if I is the initial temperature of the body, then the function $T = A + (I - A)e^{-kt}$ satisfies the condition (1). 2
- (ii) IF the initial temperature of an ingot is 1400°C and it cools in the open where the air temperature is 20°C , find the temperature after 30 minutes, given that it cooled to 1200°C in 5 minutes. 4

Section C continues next page

- c) The acceleration of a particle moving in a straight line is given by

$$\ddot{x} = -\frac{900}{x^3}$$

Where x metres is the displacement from the origin after t seconds. Initially, the particle is 10 metres to the right of the origin, moving with a velocity of 3m/s.

- (i) Find an equation for the velocity of the particle. 3
- (ii) Find the velocity of the particle when it is 100m to the right of the origin. 1
- d) The curves $y = \sin^{-1} x$ and $y = \cos^{-1} x$ intersect at point P. The acute angle between their tangents at that point is θ . Find θ to the nearest degree. You may need to use the fact that $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$, where m_1 and m_2 are the gradients of the tangent lines. 6

End of Section C.

End of Examination.

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Solutions to Section (A)

(1)

(a) $\frac{d}{dx} \left[\frac{x}{2} \ln(x^2-2) \right]$

(i) $\frac{1}{2} \ln(x^2-2) + \frac{x^2}{x^2-2}$ **2**

or $1 + \frac{1}{2} \ln(x^2-2) + \frac{2}{x^2-2}$

(ii) $\frac{d}{dx} \cos^{-1}(3x-1)$

$= \frac{-3}{\sqrt{1-(3x-1)^2}}$ **2**

OR $\frac{-3}{\sqrt{3(2x-3x^2)}}$

(b) $x = Me^{-t} + Ne^{2t}$

$\dot{x} = -Me^{-t} + 2Ne^{2t}$

$\ddot{x} = Me^{-t} + 4Ne^{2t}$

$\therefore \ddot{x} - \dot{x} - 2x$

$= Me^{-t} + 4Ne^{2t} + Me^{-t} - 2Ne^{2t}$

$= -2Me^{-t} - 2Ne^{2t}$

$= 0$

(c) $6! \times 3!$ **2**

(d) (i) $\int (4+x)^{-2} dx = \frac{-1}{4+x} + C$

(ii) $\frac{1}{2} \tan^{-1} \frac{x}{2} + C$

(iii) $\frac{1}{2} \int \frac{2x}{4+x^2} dx = \frac{\ln(4+x^2)}{2} + C$

(e) $\int_0^1 x(x-1)^5 dx$

Let $u = x-1$, $du = dx$

$x=0, u=-1. x=1, u=0.$

$\int_{-1}^0 (1+u)u^5 du$

$= \int_{-1}^0 (u^5 + u^6) du$

$= \left[\frac{u^6}{6} + \frac{u^7}{7} \right]_{-1}^0$

$= 0 - \left(\frac{1}{6} - \frac{1}{7} \right) = -\frac{1}{42}$

(f) $x^2 - 1 > 0$

$\Rightarrow x^2 > 1$

either $x > 1$
or $x < -1$

(g) > 7000 (repetition)

$\frac{7!}{2!4!4!4!}$

$= 2 \times 4^3 = 128$ **2**

(h) (i) BALLOON
(7 letters).

(i) $\frac{7!}{2!2!} = 1320$

(ii) $\frac{7!}{2!6!}$

$= 2 \times \frac{6!}{2!2!}$

$= \frac{1}{2} (6!) = 360$

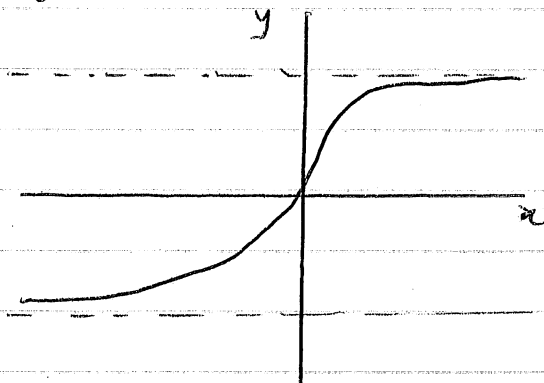
[27]

EX1 TASK 3

SECTION B.

a) i) $f(0) = 0.$

ii)



ii) Domain: $x \in \mathbb{R}.$
Range: $-\pi < y \leq \pi.$

b). i) $y = x + e^{-x}$

$$\frac{dy}{dx} = 1 - e^{-x}$$

$$1 - e^{-x} = 0$$

$$x = 0.$$

$$y = 1.$$

$$\frac{d^2y}{dx^2} = e^{-x}$$

$$\text{At } (0, 1).$$

$$e^0 > 0$$

$\therefore (0, 1)$ is a local minimum.

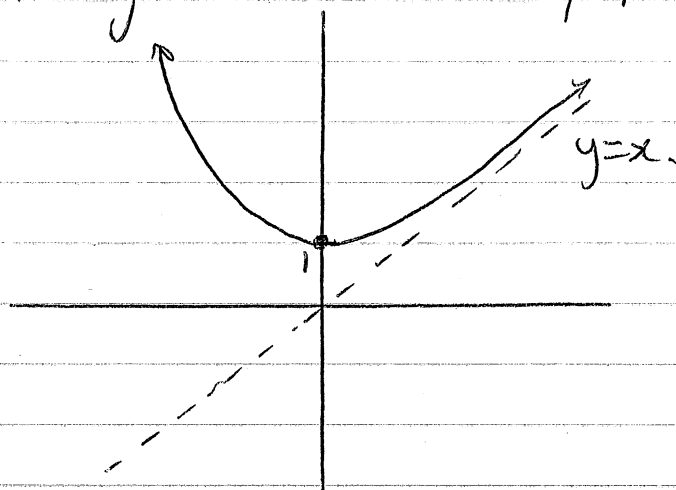
ii) $\frac{d^2y}{dx^2} = e^{-x} > 0 \quad \forall x.$

Therefore the graph is always concave up.

iii)

$$\lim_{x \rightarrow \infty} [y - x] = \lim_{x \rightarrow \infty} e^{-x} = 0.$$

$\therefore y = x$ is an asymptote.



ci). $B(0, \pi)$.

ii) $A(\frac{1}{3}, 2\pi)$
 $C(-\frac{1}{3}, 0)$.

iii). $y = \pi + 2\sin^{-1} 3x$

$$\frac{dy}{dx} = 2 \times \frac{1}{\sqrt{1-(3x)^2}} \times 3.$$

$$= \frac{6}{\sqrt{1-9x^2}}$$

At $x=0$,

$$m=6$$

$$y - y_1 = m(x - x_1).$$

$$y - \pi = 6x.$$

$$6x - y + \pi = 0.$$

d). i.)

$$\frac{r}{\sin(\frac{\pi-2\theta}{2})} = \frac{10}{\sin 2\theta}.$$

$$L = r\theta.$$

$$12 = r 2\theta$$

$$r = \frac{6}{\theta}.$$

$$\frac{r}{\cos \theta} = \frac{10}{2 \sin \theta \cos \theta}$$

$$r = \frac{5}{\sin \theta}$$

$$\therefore \frac{5}{\sin \theta} = \frac{6}{\theta}.$$

$$5\theta = 6 \sin \theta.$$

$$6 \sin \theta - 5\theta = 0.$$

ii). Let $f(\theta) = 6 \sin \theta - 5\theta$

$$f(1) = 6 \sin(1) - 5 \times 1,$$

$$\approx 0.0488$$

which is close to zero.

iii) $\theta_1 = \theta_0 - \frac{f(\theta_0)}{f'(\theta_0)}$

$$f'(\theta) = 6 \cos \theta - 5.$$

$$\theta_1 = 1 - \frac{f(1)}{f'(1)}$$

$$\approx 1.0278.$$

iv) $r = \frac{6}{\theta_1}$

$$\approx 5.84.$$

2008 Assessment #3 Mathematics Extension 1:
Solutions— Section C

3. (a)

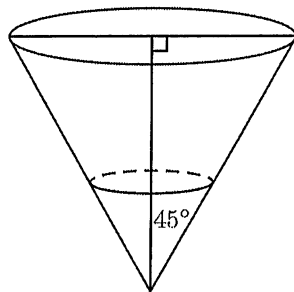


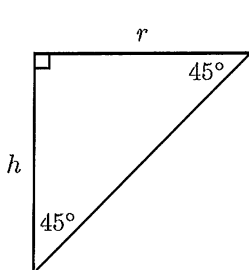
Figure not to scale.

Water is being let into the conical vessel, shown above, at a constant rate of $8 \text{ cm}^3/\text{second}$.

(i) Show that if $h \text{ cm}$ is the depth of the vessel then $r = h$ and $V = \frac{1}{3}\pi h^3$.

2

Solution:



$$\begin{aligned}\tan 45^\circ &= \frac{r}{h}, \\ &= 1.\end{aligned}$$

$$\therefore r = h.$$

{OR: As the triangle is isosceles, (base angles equal) then $r = h$.}

$$\begin{aligned}\text{Volume of cone, } V &= \frac{1}{3}\pi r^2 h, \\ &= \frac{1}{3}\pi h^3.\end{aligned}$$

When the depth is 12 cm find:

(ii) The rate of increase in the depth (in terms of π).

3

Solution: $\frac{dV}{dt} = 8 \text{ cm}^3 \text{ s}^{-1},$

$$\frac{dV}{dh} = \pi h^2,$$

$$\frac{dh}{dt} = \frac{dV}{dt} \times \frac{dh}{dV},$$

$$= \frac{8}{\pi h^2},$$

$$= \frac{1}{18\pi} \text{ cm s}^{-1} \text{ when } h = 12 \text{ cm}.$$

- (iii) The rate of increase in the area of the top surface of the water.

3

Solution: Surface area, $S = \pi h^2$,

$$\begin{aligned}\frac{dS}{dh} &= 2\pi h \\ \therefore \frac{dS}{dt} &= \frac{dS}{dh} \times \frac{dh}{dt} \\ &= 2\pi h \times \frac{8}{\pi h^2}, \\ &= \frac{16}{h}, \\ &= \frac{4}{3} \text{ cm}^2\text{s}^{-1} \text{ when } h = 12 \text{ cm.}\end{aligned}$$

- (b) The rate of decrease of temperature of a body hotter than its surroundings is proportional to the temperature difference. If A is the air temperature and T is the temperature of the body after t minutes, then

$$\frac{dT}{dt} = -k(T - A) \quad (1)$$

- (i) Show that if I is the initial temperature, then the function $T = A + (I - A)e^{-kt}$ satisfies the condition (1).

2

Solution:

$$\begin{aligned}\frac{dT}{dt} &= -k(I - A)e^{-kt}, \\ \text{but } (I - A)e^{-kt} &= T - A, \\ \therefore \frac{dT}{dt} &= -k(T - A).\end{aligned}$$

- (ii) If the initial temperature of an ingot is 1400°C and it cools in the open where the air temperature is 20°C , find the temperature after 30 minutes, given that it cooled to 1200°C in 5 minutes.

4

Solution:

$$\begin{aligned}1200 &= 20 + (1400 - 20)e^{-5k}, \\ e^{-5k} &= \frac{1180}{1380}, \\ -5k &= \ln\left(\frac{59}{69}\right), \\ k &= \frac{\ln\left(\frac{59}{69}\right)}{-5}, \\ &\approx 0.0313. \\ T_{30} &\approx 20 + (1400 - 20)e^{-30 \times 0.0313}, \\ &\approx 559.4^\circ\text{C} \text{ (4 sig. fig.)}.\end{aligned}$$

- (c) The acceleration of a particle moving in a straight line is given by

$$\ddot{x} = -\frac{900}{x^3}$$

where x metres is the displacement from the origin after t seconds. Initially, the particle is 10 metres to the right of the origin, moving with a velocity of 3 m/s.

- (i) Find an equation for the velocity of the particle.

3

Solution:

$$v \frac{dv}{dx} = -\frac{900}{x^3},$$

$$\int v dv = -900 \int x^{-3} dx,$$

$$\frac{v^2}{2} = \frac{900x^{-2}}{2} + c,$$

$$\text{i.e. } v^2 = \frac{900}{x^2} + c.$$

Initially $9 = \frac{900}{100} + c;$

$$\therefore v = \frac{30}{x} \quad (\text{positive velocity from initial conditions}).$$

- (ii) Find the velocity of the particle when it is 100 m to the right of the origin.

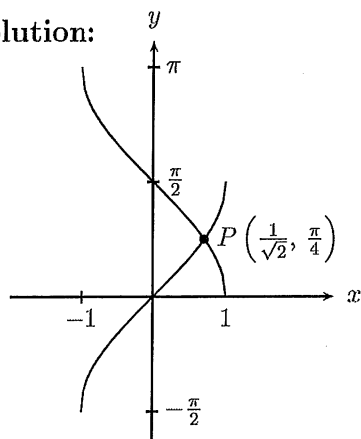
1

Solution: When $x = 100$ m, $v = 3/10 \text{ ms}^{-1}$.

- (d) The curves $y = \sin^{-1} x$ and $y = \cos^{-1} x$ intersect at point P . The acute angle between their tangents at this point is θ . Find θ to the nearest degree. You may need to use the fact that $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$, where m_1 and m_2 are the gradients of the tangent lines.

6

Solution:



From the sketch it is clear that P is $\left(\frac{1}{\sqrt{2}}, \frac{\pi}{4}\right)$.

$$\frac{d(\sin^{-1} x)}{dx} = \frac{1}{\sqrt{1-x^2}},$$

$$= \sqrt{2} \text{ at } P.$$

$$\frac{d(\cos^{-1} x)}{dx} = \frac{-1}{\sqrt{1-x^2}},$$

$$= -\sqrt{2} \text{ at } P.$$

$$\tan \theta = \left| \frac{\sqrt{2} - (-\sqrt{2})}{1 + \sqrt{2} \times (-\sqrt{2})} \right|,$$

$$= \left| \frac{2\sqrt{2}}{-1} \right|.$$

$$\therefore \theta = \tan^{-1}(2\sqrt{2}),$$

$$\approx 71^\circ \text{ (nearest degree).}$$